

Stanely Kimani Macharia

FULL-STACK SOFTWARE ENGINEER | DATABASE ADMINISTRATOR | ICT SYSTEMS ADMINISTRATOR

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CORE SKILLS

- Backend Dev — Laravel / PHP
- Database Engineering & SQL
- Mobile Dev — Flutter / Dart
- REST API Design
- Security Engineering
- DevOps & Linux
- Networking & ICT Support

TECHNOLOGIES

- Laravel 11 / 12 · PHP 8.2+
- Flutter & Dart
- PostgreSQL · MySQL
- Advanced SQL — CTEs, Self-Joins
- Filament 4.5 Admin Panel
- BLoC / Hydrated BLoC
- Tailwind CSS 4 · Vite
- Firebase (FCM, Crashlytics)
- REST APIs & Sanctum
- Git · GitHub · CI/CD
- DigitalOcean · Nginx · Linux
- DOMPDF · FPDF · QR Codes
- Africa's Talking SMS API
- Google ML Kit OCR
- AWS S3 · Laravel Horizon

SECURITY & DBA

- AES-256-CBC Encryption
- PII Data Protection
- CSP · CSRF · RBAC
- OTP / 2FA — Google Auth
- Financial Data Reconciliation
- FIFO Allocation Auditing
- Concurrency & DB Locking
- DB Backup & Recovery
- ERP Administration

INFRASTRUCTURE & SUPPORT

- Cisco Routing & Switching
- LAN / WAN / VPN
- Active Directory
- Windows Server
- ITIL Best Practices
- SLA Management
- Helpdesk L2 / L3

PROFESSIONAL SUMMARY

Results-driven Full-Stack Software Engineer, Database Administrator, and ICT Systems Administrator with 5+ years of experience designing, building, and deploying enterprise-grade web and mobile applications. Architected and delivered four production systems for Chabrin Agencies managing 500+ properties and 1,000+ tenants. Expert at solving complex technical problems at scale — from resolving systemic FIFO allocation failures across 130,000+ financial records and resyncing fractured ledger tables without data loss, to executing live zero-downtime encryption of 22,070 PII records in production. Combines deep backend and database expertise (Laravel, PostgreSQL, advanced SQL) with mobile development (Flutter, BLoC, Firebase), infrastructure management (Linux, DigitalOcean, Nginx), and enterprise security engineering (AES-256, CSP, RBAC, OTP).

EDUCATION

BSc Computer Science

Kabarak University · 2022

KCSE — Grade B

Anestor High School, Lanet · 2017

CERTIFICATIONS

- ▶ ITIL Awareness
- ▶ Cisco Networking Fundamentals
- ▶ Cybersecurity Best Practices

PROFESSIONAL EXPERIENCE

Senior ICT Systems Administrator & Software Engineer

July 2022 – Present

Chabrin Agencies Limited · Nairobi, Kenya

- ▶ **Digital Transformation:** Replaced 100% of spreadsheet-based workflows across property management, HR, payroll, and lease operations with four interconnected production systems managing 500+ properties and 1,000+ tenants.
- ▶ **Enterprise ERP:** Led full-stack development of Chab-CHIPS — Laravel 12 ERP with 52 controllers, 89 migrations, and 43 RBAC roles spanning property management, HR, statutory payroll (PAYE/NSSF/SHIF/HELB), financial returns, and a 24-table disciplinary workflow.
- ▶ **Financial Data Integrity:** Spearheaded a reconciliation project within the PROMAS ERP, identifying and resolving systemic FIFO allocation failures across 130,000+ financial records, restoring accurate tenant balances company-wide.
- ▶ **Advanced SQL & Auditing:** Wrote complex SQL scripts using CTEs and self-joins to audit overlapping API imports, isolating and eliminating cross-column duplicate financial transactions causing silent ledger corruption.
- ▶ **DB Concurrency & Recovery:** Diagnosed and resolved severe database locking issues; wrote scripts to safely resync fractured ledger tables (CHTACC) with invoice tables (CHTNTRS) with zero data loss.
- ▶ **Lease Platform:** Architected a Laravel 11 + Filament 4.5 lease system — 27-state workflow engine with multi-party digital signing (tenant → advocate → landlord) deployed on DigitalOcean at 99.9% uptime.
- ▶ **Security Engineering:** Executed live zero-downtime AES-256-CBC encryption of 22,070 PII records; implemented per-request CSP nonces, rate-limited OTP portals, and magic-byte file validation.
- ▶ **Performance:** Eliminated N+1 query patterns; added covering indexes and 5-minute auto-invalidating dashboard caching, significantly cutting admin page load times.
- ▶ **Mobile Apps:** Designed and built two Flutter apps (field officer + landlord portals) using Clean Architecture, BLoC, and Firebase with simultaneous go-live in August 2025.
- ▶ **Statutory Compliance:** Engineered full Kenya statutory compliance into the payroll engine — PAYE (graduated tax bands), NSSF (Tier I/II), SHIF, HELB, and Housing Levy — using BCMath precision arithmetic to ensure exact cent-level accuracy on every payslip.
- ▶ **Quality Assurance:** Wrote PHPUnit and Pest test suites covering critical business logic — workflow state transitions, OTP expiry, financial calculation accuracy, and RBAC permission boundaries — across 89 migrations.
- ▶ **DevOps:** Manage DigitalOcean production pipeline (PHP 8.4, Nginx, opcache); automated daily and weekly PostgreSQL backups with 30-day and 90-day retention respectively.
- ▶ **ICT Administration:** Maintain organisation-wide hardware/software infrastructure, enforce ICT security policies, manage LAN/WAN/VPN networking, and deliver staff training on digital tools and data privacy compliance.

Support Developer & Systems Specialist

Aug 2021 – July 2022

Metro ICT · Nakuru, Kenya

- ▶ Customised ERP modules to resolve workflow bottlenecks for property and retail sector clients.
- ▶ Diagnosed and resolved complex code-level issues across multiple client installations; reduced downtime through root-cause analysis over symptomatic fixes.
- ▶ Acted as L2/L3 escalation point for all infrastructure and software issues beyond first-line support scope.

IT Support Officer

Feb 2021 – July 2022

Metro ICT · Nakuru, Kenya

- ▶ Managed LAN/WAN infrastructure and resolved connectivity incidents within agreed SLA windows.
- ▶ Primary helpdesk contact for hardware and software issues; maintained asset inventory and SLA adherence records.

1. Chab-CHIPS — Enterprise Property & HR Management ERP

Laravel 12 · PostgreSQL · Laratrust RBAC · Google 2FA

- **Problem:** All property, HR, payroll, and financial operations ran on disconnected spreadsheets with no audit trail or access control.
- Built a multi-tenant ERP with 52 controllers across 9 domains — property, HR, statutory payroll, invoicing, financial returns, and a complete 24-table formal disciplinary workflow.
- Engineered multi-level maker-checker approval workflows; provisioned 43 RBAC roles with module/sub-module permission control and Google 2FA for administrators.
- Executed full MySQL → PostgreSQL migration with zero data loss using custom in-house tooling.

Laravel 12 PostgreSQL Multi-tenant ERP Payroll Automation Approval Workflows
Google 2FA PHPUnit · Pest PAYE/NSSF/SHIF/HELB Audit Logging

2. Lease Management System — Digital Signing & Document Platform

Laravel 11 · Filament 4.5 · DOMPDF · FPDF · Africa's Talking · DigitalOcean

- **Problem:** Lease execution was 100% paper-based with no verification, audit trail, or secure document storage.
- Designed a 27-state workflow engine with multi-party signing (tenant → guarantor → advocate → landlord) and 7 RBAC roles with zone-based access control.
- Built a PDF overlay engine (FPDF) for coordinate-precise signature stamping and a visual calibration artisan command (`lease:visual-map`).
- OTP-secured signing portal — rate-limited (5 OTP/min, 3 signatures/min), signed URL expiry, device fingerprinting, geolocation capture.
- Live zero-downtime AES-256-CBC encryption of 22,070 PII records; per-request CSP nonces; magic-byte file validation.
- **Impact:** 500+ properties fully digital end-to-end; live at leases-docs.chabrinagencies.com, 99.9% uptime.

Digital Signing OTP / SMS PDF Overlay AES-256 CSP BCMath Financials DigitalOcean

3. Chabrin Field Officer Mobile App

Flutter · BLoC · Drift ORM · Google ML Kit · Firebase · Clean Architecture

- **Problem:** Field officers managing on-site properties had no mobile access to tenant, payment, or inspection data.
- Cross-platform Flutter app (Android, iOS, Web) — Clean Architecture, BLoC state management, dependency injection (`get_it/injectable`).
- Offline-first strategy using Drift SQLite ORM — fully functional without connectivity, auto-syncs when online.
- Google ML Kit OCR — field officers scan tenant ID documents in-app, eliminating manual data entry errors during inspections.
- v2.4-alpha · 161 commits · 16 modules: properties, tenants, payments, inspections, leave, payslips, documents.

Flutter BLoC Clean Architecture Offline-First ML Kit OCR Firebase

4. Chabrin Landlord Self-Service Portal

Flutter · BLoC · Firebase · responsive_framework · Mason · Custom Charts

- **Problem:** Landlords had zero visibility into portfolio or financials without requesting manual reports from Chabrin staff.
- Flutter mobile + web app delivering real-time portfolio view, rental income statements, agency deductions, and analytics dashboard.
- Built custom `charts_flutter` and shared `local_storage` packages reused across the Chabrin Flutter app suite.
- Single responsive codebase (mobile + desktop); Mason code generation for consistent Clean Architecture feature scaffolding.
- Coordinated simultaneous go-live with Field Officer app (August 2025), managing cross-app API compatibility.

Flutter Responsive Web Custom Packages Data Visualisation Mason Self-Service Portal